

Matrix Methods in Machine Learning

Lecture 2: Linear Independence and Rank

Zamir Hussain
Department of Mathematics
University of Wah

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📅 Week 2

Topic: Linear Independence and Rank

Subtopics: Spanning vectors, linear combinations, Rank, linear independence, matrix properties, and ML Applications

🔑 Why This Matters for Math Students

Understanding linear independence and rank helps you detect redundancy in data, determine the solvability of equations, and form the foundation for machine learning methods like regression and dimensionality reduction.

1 Spanning Vectors and Linear Combinations

💡 Theoretical Foundations

A **linear combination** of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is any vector of the form:

$$\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n,$$

where $c_1, c_2, \dots, c_n$ are real numbers (called scalars).

The **span** of $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is the set of all possible linear combinations:

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \{c_1\mathbf{v}_1 + \dots + c_n\mathbf{v}_n \mid c_i \in \mathbb{R}\}.$$

Example 1: Linear Combination in $\mathbb{R}^2$

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then any vector

$$\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

Hence, $\mathbf{v}_1$ and $\mathbf{v}_2$ **span** the entire $\mathbb{R}^2$ plane.

Example 2: Dependent Span in $\mathbb{R}^3$

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}.$$

Since $\mathbf{v}_2 = 2\mathbf{v}_1$, all combinations lie along one line. Thus, the span is **one-dimensional**.

2 Linear Independence

💡 Theoretical Foundations

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are **linearly independent** if

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0} \Rightarrow c_1 = c_2 = \dots = c_n = 0.$$

Otherwise, they are **dependent**.

Example 3: Independent Vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{0} \Rightarrow c_1 = c_2 = 0.$$

Hence, independent.

Example 4: Dependent Vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{0} \Rightarrow c_1 = -2c_2.$$

Infinite solutions $\Rightarrow$ dependent.

3 Matrix Form and Rank

We can represent a set of vectors as columns of a matrix:

$$A = \begin{bmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{bmatrix}.$$

The **rank** of A is the number of linearly independent columns (or rows).

Example 5: Finding Rank by Row Reduction

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Only one nonzero row $\Rightarrow \text{rank}(A) = 1$.

Example 6: Full Rank Matrix

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Upper triangular with 3 pivots $\Rightarrow \text{rank}(A) = 3$.

4 Connection Between Rank and Independence

Rank of A = Number of independent columns
Rank = $n \Rightarrow$ Columns are independent
Rank < $n \Rightarrow$ Columns are dependent

Machine Learning Applications

Applications in Machine Learning:

1. **Feature Redundancy:** Dependent features provide no new information; PCA removes them.

2. **Linear Regression:** For the normal equation

$$\beta = (X^T X)^{-1} X^T y,$$

X must have full column rank for β to be unique.

3. **Neural Networks:** Redundant neurons or weights make matrices rank-deficient, reducing model expressiveness.

5 Practice Problem

Determine if the following vectors are linearly independent:

$$\mathbf{v}_1 = (1, 1, 0), \quad \mathbf{v}_2 = (0, 1, 1), \quad \mathbf{v}_3 = (1, 2, 1).$$

Set up

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 = \mathbf{0} \Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \mathbf{0}.$$

Row reduction gives 3 pivots $\Rightarrow$ only zero solution $\Rightarrow$ Independent.

Vectors are linearly independent.

6 Summary

Concept	Meaning	Example
Span	All linear combinations	$\text{Span}\{(1, 0), (0, 1)\} = \mathbb{R}^2$
Linear Independence	No vector is a combo of others	$(1, 0), (0, 1)$ independent
Dependence	One vector = combo of others	$(1, 2), (2, 4)$ dependent
Rank	Number of independent columns	$\text{rank} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = 1$

Career Pathways & Future Scope

A strong understanding of rank and linear independence leads to insight into:

- Principal Component Analysis (PCA)
- Matrix Factorization
- Dimensionality Reduction
- Linear Regression and Neural Networks